

Обратная матрица.

Кочеткова М.П., 2.2

Метод присоединенной матрицы

5° 1.4.57.

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0,5 & 0 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0,5 & 0 \end{vmatrix} = 0 + 0 + 0 - 0 - 0 + 1 \cdot 0,5 \cdot 2 = 1 \neq 0$$

$$2) A_{11} = (-1)^2 \cdot \begin{vmatrix} 0 & 2 \\ 0,5 & 0 \end{vmatrix} = 1 \cdot (0 - 2 \cdot 0,5) = -1$$

$$A_{12} = (-1)^3 \cdot \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{23} = (-1)^5 \cdot \begin{vmatrix} -1 & 0 \\ 0 & 0,5 \end{vmatrix} = -1 \cdot (-0,5) = 0,5$$

$$A_{13} = (-1)^4 \cdot \begin{vmatrix} 0 & 0 \\ 0 & 0,5 \end{vmatrix} = 0$$

$$A_{31} = (-1)^4 \cdot \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0$$

$$A_{21} = (-1)^3 \cdot \begin{vmatrix} 0 & 0 \\ 0,5 & 0 \end{vmatrix} = 0$$

$$A_{32} = (-1)^5 \cdot \begin{vmatrix} -1 & 0 \\ 0 & 2 \end{vmatrix} = -1 \cdot (-2) = 2$$

$$A_{22} = (-1)^4 \cdot \begin{vmatrix} -1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{33} = (-1)^6 \cdot \begin{vmatrix} -1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$3) A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0,5 \\ 0 & 2 & 0 \end{pmatrix}$$

$$4) \tilde{A} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0,5 & 0 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{1} \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0,5 & 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0,5 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0,5 & 0 \end{pmatrix}$$

5) 1.4.58.

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 8 & 3 & -6 \\ -4 & -1 & 3 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 1 & -1 \\ 8 & 3 & -6 \\ -4 & -1 & 3 \end{vmatrix} = 9 + 8 + 24 - 12 - 24 - 6 = -1 \neq 0$$

$$2) A_{11} = (-1)^1 \cdot \begin{vmatrix} 3 & -6 \\ -1 & 3 \end{vmatrix} = 1 \cdot (9 - 6) = 3$$

$$A_{12} = (-1)^2 \cdot \begin{vmatrix} 8 & -6 \\ -4 & 3 \end{vmatrix} = -1 \cdot (24 - 24) = 0$$

$$A_{13} = (-1)^3 \cdot \begin{vmatrix} 8 & 3 \\ -4 & -1 \end{vmatrix} = 1 \cdot (-8 + 12) = 4$$

$$A_{21} = (-1)^3 \cdot \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = -1 \cdot (3 - 1) = -2$$

$$A_{22} = (-1)^4 \cdot \begin{vmatrix} 1 & -1 \\ -4 & 3 \end{vmatrix} = 1 \cdot (3 - 4) = -1$$

$$A_{23} = (-1)^5 \cdot \begin{vmatrix} 1 & 1 \\ -4 & -1 \end{vmatrix} = -1 \cdot (-1 + 4) = -3$$

$$A_{31} = (-1)^4 \cdot \begin{vmatrix} 1 & -1 \\ 3 & -6 \end{vmatrix} = 1 \cdot (-6 + 3) = -3$$

$$A_{32} = (-1)^5 \cdot \begin{vmatrix} 1 & -1 \\ 8 & -6 \end{vmatrix} = -1 \cdot (-6 + 8) = -2$$

$$A_{33} = (-1)^6 \cdot \begin{vmatrix} 1 & 1 \\ 8 & 3 \end{vmatrix} = 1 \cdot (3 - 8) = -5$$

$$3) A = \begin{pmatrix} 3 & 0 & 4 \\ -2 & -1 & -3 \\ -3 & -2 & -5 \end{pmatrix}$$

$$4) \hat{A} = \begin{pmatrix} 3 & -2 & -3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{-1} \cdot \begin{pmatrix} 3 & -2 & -3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix} = -1 \cdot \begin{pmatrix} 3 & -2 & -3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix} = \begin{pmatrix} -3 & 2 & 3 \\ 0 & 1 & 2 \\ -4 & 3 & 5 \end{pmatrix}$$

5.1.4.39

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{vmatrix} = -4 + 4 + 8 + 8 - 2 - 8 = 6 \neq 0$$

$$c) A_{11} = (-1)^{2+2} \cdot \begin{vmatrix} -1 & 2 \\ 1 & 4 \end{vmatrix} = 1 \cdot (-4 - 2) = -6$$

$$A_{12} = (-1)^{5+2} \cdot \begin{vmatrix} 2 & 2 \\ 4 & 4 \end{vmatrix} = -1 \cdot (8 - 8) = 0$$

$$A_{13} = (-1)^{5+3} \cdot \begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix} = 1 \cdot (2 + 4) = 6$$

$$A_{21} = (-1)^{3+1} \cdot \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = -1 \cdot (4 - 2) = -2$$

$$A_{22} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & 2 \\ 4 & 4 \end{vmatrix} = 1 \cdot (4 - 8) = -4$$

$$A_{23} = (-1)^{3+3} \cdot \begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = -1 \cdot (1 - 4) = 3$$

$$A_{31} = (-1)^{4+1} \cdot \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} = 1 \cdot (2 + 2) = 4$$

$$A_{32} = (-1)^{4+2} \cdot \begin{vmatrix} 1 & 2 \\ 2 & 2 \end{vmatrix} = -1 \cdot (2 - 4) = 2$$

$$A_{33} = (-1)^{4+3} \cdot \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = 1 \cdot (-1 - 2) = -3$$

$$3) A = \begin{pmatrix} -6 & 0 & 6 \\ -2 & -4 & 3 \\ 4 & 2 & -3 \end{pmatrix}$$

$$4) \tilde{A} = \begin{pmatrix} -6 & -2 & 4 \\ 0 & -4 & 2 \\ 6 & 3 & -3 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{6} \cdot \begin{pmatrix} -6 & -2 & 4 \\ 0 & -4 & 2 \\ 6 & 3 & -3 \end{pmatrix} = \begin{pmatrix} -1 & -\frac{1}{3} & \frac{2}{3} \\ 0 & -\frac{2}{3} & \frac{1}{3} \\ 1 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

5.1.4.40

$$A = \begin{pmatrix} 3 & 4 & 2 \\ 2 & -4 & -3 \\ 1 & 5 & 1 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 3 & 4 & 2 \\ 2 & -4 & -3 \\ 1 & 5 & 1 \end{vmatrix} = -12 + 20 - 12 + 8 + 45 - 8 = 41$$

$$2) A_{11} = (-1)^2 \cdot \begin{vmatrix} -4 & -3 \\ 5 & 1 \end{vmatrix} = 1 \cdot (-4 + 15) = 11$$

$$A_{23} = (-1)^5 \cdot \begin{vmatrix} 3 & 4 \\ 1 & 5 \end{vmatrix} = -1 \cdot (15 - 4) = -11$$

$$A_{12} = (-1)^3 \cdot \begin{vmatrix} 2 & -3 \\ 1 & 1 \end{vmatrix} = -1 \cdot (2 + 3) = -5$$

$$A_{31} = (-1)^4 \cdot \begin{vmatrix} 4 & 2 \\ -4 & -3 \end{vmatrix} = 1 \cdot (-12 + 8) = -4$$

$$A_{13} = (-1)^4 \cdot \begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} = 1 \cdot (10 - 4) = 14$$

$$A_{32} = (-1)^5 \cdot \begin{vmatrix} 3 & 2 \\ 2 & -3 \end{vmatrix} = -1 \cdot (-9 - 4) = 13$$

$$A_{21} = (-1)^5 \cdot \begin{vmatrix} 4 & 2 \\ 5 & 1 \end{vmatrix} = -1 \cdot (4 - 10) = 6$$

$$A_{33} = (-1)^6 \cdot \begin{vmatrix} 3 & 4 \\ 2 & -4 \end{vmatrix} = 1 \cdot (-12 - 8) = -20$$

$$A_{22} = (-1)^6 \cdot \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = 1 \cdot (3 - 2) = 1$$

$$3) A = \begin{pmatrix} 11 & -5 & 14 \\ 6 & 1 & -11 \\ -4 & 13 & -20 \end{pmatrix}$$

$$4) \tilde{A} = \begin{pmatrix} 11 & 6 & -4 \\ -5 & 1 & 13 \\ 14 & -11 & -20 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{41} \cdot \begin{pmatrix} 11 & 6 & -4 \\ -5 & 1 & 13 \\ 14 & -11 & -20 \end{pmatrix} = \begin{pmatrix} \frac{11}{41} & \frac{6}{41} & -\frac{4}{41} \\ -\frac{5}{41} & \frac{1}{41} & \frac{13}{41} \\ \frac{14}{41} & -\frac{11}{41} & -\frac{20}{41} \end{pmatrix}$$

Зад. 1. 4. 41

$$A = \begin{pmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{vmatrix} = 0 \cdot 24 - 0 \cdot 4 = 0 + 24 - 3 + 6 - 27 + 0 = 0 \Rightarrow$$

$\Rightarrow$ обратной матрицы не существует

Зад. 1. 4. 42

$$A = \begin{pmatrix} 5 & 8 & -1 \\ 2 & -3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 5 & 8 & -1 \\ 2 & -3 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -45 - 4 + 16 - 3 - 20 - 48 = -104$$

$$2) A_{11} = (-1)^{2+2} \cdot \begin{vmatrix} -3 & 2 \\ 2 & 3 \end{vmatrix} = 1 \cdot (-9 - 4) = -13$$

$$A_{25} = (-1)^{5+5} \cdot \begin{vmatrix} 5 & 8 \\ 1 & 2 \end{vmatrix} = 1 \cdot (10 - 8) = 2$$

$$A_{12} = (-1)^{5+3} \cdot \begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} = -1 \cdot (6 - 2) = -4$$

$$A_{41} = (-1)^{4+1} \cdot \begin{vmatrix} 8 & -1 \\ -3 & 2 \end{vmatrix} = -1 \cdot (16 - 3) = -13$$

$$A_{13} = (-1)^{4+3} \cdot \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = -1 \cdot (4 + 3) = -7$$

$$A_{32} = (-1)^{5+2} \cdot \begin{vmatrix} 5 & -1 \\ 2 & 2 \end{vmatrix} = -1 \cdot (10 + 2) = -12$$

$$A_{21} = (-1)^{5+1} \cdot \begin{vmatrix} 8 & -1 \\ 2 & 3 \end{vmatrix} = -1 \cdot (24 + 2) = -26$$

$$A_{33} = (-1)^{6+3} \cdot \begin{vmatrix} 5 & 8 \\ 2 & -3 \end{vmatrix} = -1 \cdot (-15 - 16) = -31$$

$$A_{22} = (-1)^{4+2} \cdot \begin{vmatrix} 5 & -1 \\ 1 & 3 \end{vmatrix} = 1 \cdot (15 + 1) = 16$$

$$3) A = \begin{pmatrix} -13 & -4 & 7 \\ -26 & 16 & -2 \\ 13 & -12 & -31 \end{pmatrix}$$

$$4) \tilde{A} = \begin{pmatrix} -13 & -26 & 13 \\ -4 & 16 & -12 \\ 7 & -2 & -31 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{-104} \cdot \begin{pmatrix} -13 & -26 & 13 \\ -4 & 16 & -12 \\ 7 & -2 & -31 \end{pmatrix} = \begin{pmatrix} \frac{13}{104} & \frac{26}{104} & \frac{-13}{104} \\ \frac{4}{104} & \frac{-16}{104} & \frac{12}{104} \\ \frac{-7}{104} & \frac{2}{104} & \frac{31}{104} \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & \frac{1}{4} & \frac{-1}{8} \\ \frac{1}{26} & \frac{-1}{26} & \frac{3}{26} \\ \frac{-7}{104} & \frac{1}{52} & \frac{31}{104} \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & \frac{1}{4} & \frac{-1}{8} \\ \frac{1}{26} & \frac{-1}{26} & \frac{3}{26} \\ \frac{-7}{104} & \frac{1}{52} & \frac{31}{104} \end{pmatrix}$$

Метод элементарных преобразований

1.4.48

$$A = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{vmatrix} = 4 + 1 + 1 - 2 - 2 - 1 = 1 \neq 0$$

$$2) \Gamma = (A|E) = \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 & 1 & 0 \\ -1 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \text{II} + \text{I} \\ \text{III} + \text{I} \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right)$$

$$3) \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right) \begin{array}{l} \text{I} + \text{II} + \text{III} \\ \end{array} \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right) = (E|A^{-1}) \Rightarrow$$

$$\Rightarrow A^{-1} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

1.4.44

$$A = \begin{pmatrix} 2 & 7 & 3 \\ 3 & 9 & 4 \\ 1 & 5 & 8 \end{pmatrix}$$

St 1.4.44

$$A = \begin{pmatrix} 2 & 7 & 3 \\ 3 & 9 & 4 \\ 1 & 5 & 3 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 2 & 7 & 3 \\ 3 & 9 & 4 \\ 1 & 5 & 3 \end{vmatrix} = 54 + 45 + 28 - 27 - 40 - 63 = -3$$

$$2) \tilde{A} = (A|E) = \left(\begin{array}{ccc|ccc} 2 & 7 & 3 & 1 & 0 & 0 \\ 3 & 9 & 4 & 0 & 1 & 0 \\ 1 & 5 & 3 & 0 & 0 & 1 \end{array} \right)$$

$$3) \left(\begin{array}{ccc|ccc} 2 & 7 & 3 & 1 & 0 & 0 \\ 3 & 9 & 4 & 0 & 1 & 0 \\ 1 & 5 & 3 & 0 & 0 & 1 \end{array} \right) \xrightarrow{I \leftrightarrow III} \sim \left(\begin{array}{ccc|ccc} 1 & 5 & 3 & 0 & 0 & 1 \\ 3 & 9 & 4 & 0 & 1 & 0 \\ 2 & 7 & 3 & 1 & 0 & 0 \end{array} \right) \xrightarrow{\begin{array}{l} II - 3I \\ III - 2I \end{array}} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 5 & 3 & 0 & 0 & 1 \\ 0 & -6 & -5 & 0 & 1 & -3 \\ 0 & -3 & -3 & 1 & 0 & -2 \end{array} \right) \xrightarrow{II \cdot (-1/6)} \sim \left(\begin{array}{ccc|ccc} 1 & 5 & 3 & 0 & 0 & 1 \\ 0 & 1 & 5/6 & 0 & 1/6 & 1/2 \\ 0 & -3 & -3 & 1 & 0 & -2 \end{array} \right) \xrightarrow{III + 3II} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1/6 & 0 & 5/6 & -1/2 \\ 0 & 1 & 5/6 & 0 & 1/6 & 1/2 \\ 0 & 0 & -1/2 & 1 & -1/2 & -1/2 \end{array} \right) \xrightarrow{III \cdot (-2)} \sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1/6 & 0 & 5/6 & -1/2 \\ 0 & 1 & 5/6 & 0 & 1/6 & 1/2 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array} \right) \xrightarrow{\begin{array}{l} I + 1/6 III \\ II - 5/6 III \end{array}} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/3 & 2 & -1/3 \\ 0 & 1 & 0 & 5/3 & -1 & -1/3 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array} \right) = (E|A^{-1}) \Rightarrow$$

$$\Rightarrow A^{-1} = \begin{pmatrix} -1/3 & 2 & -1/3 \\ 5/3 & -1 & -1/3 \\ -2 & 1 & 1 \end{pmatrix}$$

SG 1.4.45

$$A = \begin{pmatrix} 1 & 2 & -2 & 4 \\ 2 & 6 & 1 & 0 \\ 3 & 0 & 1 & 2 \\ -1 & 4 & 5 & -4 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 2 & -2 & 4 \\ 2 & 6 & 1 & 0 \\ 3 & 0 & 1 & 2 \\ -1 & 4 & 5 & -4 \end{vmatrix} = \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix} = 1 \cdot \begin{vmatrix} 6 & 1 & 0 \\ 0 & 1 & 2 \\ 4 & 5 & -4 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & -2 & 4 \\ 0 & 1 & 2 \\ 4 & 5 & -4 \end{vmatrix} + 2 \cdot \begin{vmatrix} 2 & -2 & 4 \\ 6 & 1 & 0 \\ 4 & 5 & -4 \end{vmatrix} - 4 \cdot \begin{vmatrix} 2 & -2 & 4 \\ 3 & 0 & 1 \\ -1 & 4 & 5 \end{vmatrix}$$

$$+ 1 \cdot \begin{vmatrix} 2 & -2 & 4 \\ 6 & 1 & 0 \\ 0 & 1 & 2 \end{vmatrix} = (-24 + 8 + 0 - 0 - 60 - 0) - 2(-8 - 16 + 0 - 16 - 20 - 0) + 2(-8 + 120 - 0 - 16 - 0 - 48) +$$

$$+ (4 + 24 + 0 - 0 - 0 + 24) = -76 - 2(-60) + 2(48) + 52 = 240 \neq 0$$

$$2) \Gamma = (A|E) = \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 2 & 6 & 1 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ -1 & 4 & 5 & -4 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$3) \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 2 & 6 & 1 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ -1 & 4 & 5 & -4 & 0 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \\ R-2I \\ III-3I \\ IV+I \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 0 & 2 & 5 & -8 & -2 & 1 & 0 & 0 \\ 0 & -6 & 7 & -10 & -3 & 0 & 1 & 0 \\ 0 & 6 & 3 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \\ R \cdot \frac{1}{2} \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & -4 & -1 & \frac{1}{2} & 0 & 0 \\ 0 & -6 & 7 & -10 & -3 & 0 & 1 & 0 \\ 0 & 6 & 3 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} I-2II \\ \\ III+6II \\ IV-6II \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & -7 & 12 & 3 & -1 & 0 & 0 \\ 0 & 1 & 5/2 & -4 & -1 & 1/2 & 0 & 0 \\ 0 & 0 & 22 & -34 & -9 & 5 & 1 & 0 \\ 0 & 0 & -12 & 24 & 7 & -3 & 0 & 1 \end{array} \right) \text{III} \cdot 1/22 \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & -7 & 12 & 3 & -1 & 0 & 0 \\ 0 & 1 & 5/2 & -4 & -1 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & -12/11 & -9/22 & 3/22 & 1/22 & 0 \\ 0 & 0 & -12 & 24 & 7 & -3 & 0 & 1 \end{array} \right) \begin{array}{l} \text{I} + 7\text{III} \\ \text{II} - (5/2)\text{III} \\ \text{IV} + 12\text{III} \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 13/11 & 3/22 & -1/22 & 7/22 & 0 \\ 0 & 1 & 0 & -3/22 & 1/44 & 7/44 & -5/44 & 0 \\ 0 & 0 & 1 & -12/11 & -9/22 & 3/22 & 1/22 & 0 \\ 0 & 0 & 0 & 60/11 & 23/11 & -15/11 & 5/11 & 1 \end{array} \right) \text{IV} \cdot 1/60 \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 13/11 & 3/22 & -1/22 & 7/22 & 0 \\ 0 & 1 & 0 & -3/22 & 1/44 & 7/44 & -5/44 & 0 \\ 0 & 0 & 1 & -12/11 & -9/22 & 3/22 & 1/22 & 0 \\ 0 & 0 & 0 & 1 & 23/60 & -1/4 & 1/10 & 1/60 \end{array} \right) \begin{array}{l} \text{I} - (13/11)\text{IV} \\ \text{II} + (3/22)\text{IV} \\ \text{III} + (12/11)\text{IV} \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -19/60 & 1/4 & 1/5 & -13/60 \\ 0 & 1 & 0 & 0 & 3/40 & 1/8 & -1/10 & 1/40 \\ 0 & 0 & 1 & 0 & 1/60 & -1/4 & 1/5 & 17/60 \\ 0 & 0 & 0 & 1 & 23/60 & -1/4 & 1/10 & 1/60 \end{array} \right) = (E|A^{-1}) \Rightarrow$$

$$\Rightarrow A^{-1} = \begin{pmatrix} -19/60 & 1/4 & 1/5 & -13/60 \\ 3/40 & 1/8 & -1/10 & 1/40 \\ 1/60 & -1/4 & 1/5 & 17/60 \\ 23/60 & -1/4 & 1/10 & 1/60 \end{pmatrix}$$

Матричные уравнения.

§1.4.50

$$X \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 - 6 = -2 \neq 0$$

$$2) A_{11} = (-1)^1 \cdot |4| = -4 \quad A_{12} = (-1)^2 \cdot |2| = 2$$

$$A_{21} = (-1)^3 \cdot |3| = -3 \quad A_{22} = (-1)^4 \cdot |1| = 1$$

$$3) A = \begin{pmatrix} 4 & -5 \\ -2 & 1 \end{pmatrix} \quad 4) \tilde{A} = \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{-1} \cdot \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$$

$$6) X = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

§1.4.51

$$X \cdot \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 4 & 3 \\ -5 & -4 \end{vmatrix} = -16 + 15 = -1 \neq 0$$

$$2) A^{-1} = \frac{1}{-1} \cdot \begin{pmatrix} -4 & -3 \\ 5 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix}$$

$$3) X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix}$$

№ 1.4.52

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1 - 1 = 0 \Rightarrow$$

$\Rightarrow$ обратной матрицы не существует

№ 1.4.53

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1 - 1 = 0 \Rightarrow$$

$\Rightarrow$ обратной матрицы не существует

№ 1.4.54

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot X \cdot \begin{pmatrix} -5 & -6 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} -5 & -6 \\ -4 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 3 + 2 = 5 \neq 0 \Rightarrow A^{-1} = \frac{1}{5} \cdot \begin{pmatrix} 3 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 3/5 & 1/5 \\ -2/5 & 1/5 \end{pmatrix}$$

$$\det C = \begin{vmatrix} -5 & -6 \\ -4 & 5 \end{vmatrix} = -25 - 24 = -49 \neq 0 \Rightarrow C^{-1} = \frac{1}{-49} \cdot \begin{pmatrix} 5 & 6 \\ 4 & -5 \end{pmatrix} = \begin{pmatrix} -5/49 & -6/49 \\ 4/49 & 5/49 \end{pmatrix}$$

$$2) X = A^{-1} \cdot B \cdot C^{-1} = \begin{pmatrix} 3/5 & 1/5 \\ -2/5 & 1/5 \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} -5/49 & -6/49 \\ 4/49 & 5/49 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} -5/49 & -6/49 \\ 4/49 & 5/49 \end{pmatrix} = \begin{pmatrix} -5/49 & -6/49 \\ 4/49 & 5/49 \end{pmatrix}$$

§ 1.4.55

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot X \cdot \begin{pmatrix} 2 & -2 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, C = \begin{pmatrix} 2 & -2 \\ -4 & 5 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 3 + 2 = 5 \neq 0 \Rightarrow A^{-1} = \frac{1}{5} \cdot \begin{pmatrix} 3 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix}$$

$$\det C = \begin{vmatrix} 2 & -2 \\ -4 & 5 \end{vmatrix} = 10 - 8 = 2 \neq 0 \Rightarrow C^{-1} = \frac{1}{2} \cdot \begin{pmatrix} 5 & 2 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix}$$

$$2) X = A^{-1} \cdot B \cdot C^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix}$$

§ 1.4.56

$$X \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 6 \neq 0$$

$$2) A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 1 \cdot 6 = 6$$

$$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = -1 \cdot 0 = 0$$

$$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = -1 \cdot 0 = 0$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 0 & 0 \\ 2 & 0 \end{vmatrix} = 1 \cdot 0 = 0$$

$$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 1 \cdot 0 = 0$$

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = -1 \cdot 0 = 0$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = 1 \cdot 0 = 0$$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = 1 \cdot 2 = 2$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix} = 1 \cdot 3 = 3$$

$$3) A = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$4) \tilde{A} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$5) A^{-1} = \frac{1}{6} \cdot \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$6) X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 4 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

§ 1.4.57

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix}, B = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{vmatrix} = 3 - 12 + 0 - 0 - 2 + 4 = -7 \neq 0$$

$$2) A_{11} = (-1)^2 \cdot \begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} = 1 \cdot 1 = 1$$

$$A_{23} = (-1)^5 \cdot \begin{vmatrix} 1 & -2 \\ 0 & -2 \end{vmatrix} = -1 \cdot (-2) = 2$$

$$A_{12} = (-1)^3 \cdot \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = -1 \cdot 2 = -2$$

$$A_{31} = (-1)^4 \cdot \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} = 1 \cdot (-7) = -7$$

$$A_{13} = (-1)^4 \cdot \begin{vmatrix} 2 & 3 \\ 0 & -2 \end{vmatrix} = 1 \cdot (-4) = -4$$

$$A_{32} = (-1)^5 \cdot \begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = -1 \cdot (-7) = 7$$

$$A_{21} = (-1)^5 \cdot \begin{vmatrix} -2 & 3 \\ -2 & 1 \end{vmatrix} = -1 \cdot 4 = -4$$

$$A_{33} = (-1)^6 \cdot \begin{vmatrix} 1 & -2 \\ 2 & 3 \end{vmatrix} = 1 \cdot 7 = 7$$

$$A_{22} = (-1)^4 \cdot \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 = 1$$

$$3) A = \begin{pmatrix} 1 & -2 & -4 \\ -4 & 1 & 2 \\ -7 & 7 & 7 \end{pmatrix}$$

$$4) \tilde{A} = \begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix}$$

$$5) A^{-1} = -\frac{1}{7} \cdot \begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix} = \begin{pmatrix} 1/7 & 4/7 & 1 \\ 2/7 & -1/7 & -1 \\ 4/7 & -2/7 & -1 \end{pmatrix}$$

$$6) X = A^{-1} \cdot B = \begin{pmatrix} 1/7 & 4/7 & 1 \\ 2/7 & -1/7 & -1 \\ 4/7 & -2/7 & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 15/7 \\ -16/7 \\ -11/7 \end{pmatrix}$$

№ 1.4.58

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix} \cdot X \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$$

$$1) \det A = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{vmatrix} = 3 - 12 + 0 - 0 - 2 + 4 = -7 \neq 0$$

$$\det C = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{vmatrix} = 0 + 84 + 86 - 105 - 48 - 0 = 27 \neq 0$$

$$2) A^{-1} - \text{CM. } \text{№ 1.4.57} \Rightarrow A^{-1} = \begin{pmatrix} 1/7 & 4/7 & 1 \\ 2/7 & -1/7 & -1 \\ 4/7 & -2/7 & -1 \end{pmatrix}$$

$$3) C_{11} = (-1)^2 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 0 \end{vmatrix} = 1 \cdot (-48) = -48$$

$$C_{23} = (-1)^5 \cdot \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = -1 \cdot (-6) = 6$$

$$C_{12} = (-1)^3 \cdot \begin{vmatrix} 4 & 6 \\ 7 & 0 \end{vmatrix} = -1 \cdot (-42) = 42$$

$$C_{31} = (-1)^9 \cdot \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = -1 \cdot (-3) = 3$$

$$C_{13} = (-1)^4 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = 1 \cdot (-3) = -3$$

$$C_{32} = (-1)^5 \cdot \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = -1 \cdot (-6) = 6$$

$$C_{21} = (-1)^5 \cdot \begin{vmatrix} 2 & 3 \\ 8 & 0 \end{vmatrix} = -1 \cdot (-24) = 24$$

$$C_{33} = (-1)^6 \cdot \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = 1 \cdot (-3) = -3$$

$$C_{22} = (-1)^4 \cdot \begin{vmatrix} 1 & 3 \\ 7 & 0 \end{vmatrix} = 1 \cdot (-21) = -21$$

$$4) \quad C = \begin{pmatrix} -48 & 42 & -3 \\ 24 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix}$$

$$5) \quad \tilde{C} = \begin{pmatrix} -48 & 24 & -3 \\ 42 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix}$$

$$6) \quad C^{-1} = \frac{1}{27} \cdot \begin{pmatrix} -48 & 24 & -3 \\ 42 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix} = \begin{pmatrix} -16/9 & 8/9 & -1/9 \\ 14/9 & -7/9 & 2/9 \\ -1/9 & 2/9 & -1/9 \end{pmatrix}$$

$$7) \quad X = A^{-1} \cdot B \cdot C^{-1} = \begin{pmatrix} -1/7 & 4/7 & 1 \\ 2/7 & -1/7 & -1 \\ 4/7 & -2/7 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} \cdot \begin{pmatrix} -16/9 & 8/9 & -1/9 \\ 14/9 & -7/9 & 2/9 \\ -1/9 & 2/9 & -1/9 \end{pmatrix} =$$

$$= \begin{pmatrix} 64/7 & 74/7 & 3 \\ -51/7 & -57/7 & 0 \\ -53/7 & -58/7 & 0 \end{pmatrix} \cdot \begin{pmatrix} -16/9 & 8/9 & -1/9 \\ 14/9 & -7/9 & 2/9 \\ -1/9 & 2/9 & -1/9 \end{pmatrix} = \begin{pmatrix} -1/7 & 4/7 & 1 \\ 2/7 & -1/7 & -1 \\ 4/7 & -2/7 & -1 \end{pmatrix}$$

$\Rightarrow$